

1. What is normalisation?

Normalization is a step-by-step process of restructuring tables to minimize data anomalies.

2. When is a table in 1NF?

The table is in the first normal form stage when there are no repeating groups, so every cell has only one value. There's a primary key that's been defined and all attributes depend on that primary key.

3. When is a table in 2NF?

The table is in the second normal form stage, when first of all it has already been in the first normal form stage, and there are no partial dependencies.

This is possible when you have a composite key, so a primary key that's been made of two columns. Essentially every other column must depend on both parts of the key, not just one.

4. When is a table in 3NF?

A table is in the third normal form stage when it has been in the second normal form stage and it has no transitive dependencies. This means that a non-key column should not depend on another non-key column. For example, the price depends on the product ID, not on the customer name.

5.

The diagram illustrates functional dependencies using red and blue arrows on the table above. Red arrows indicate dependencies on the primary key (INV_NUM, PROD_NUM), while blue arrows indicate dependencies on other attributes.

INV_NUM	PROD_NUM	SALE_DATE	PROD_LABEL	VEND_CODE	VEND_NAME	QUANT_SOLD	PROD_PRICE
211347	AA-E3522QW	15-Jan-2018	Rotary sander	211	NeverFail, Inc.	1	\$34.46
211347	QD-300932X	15-Jan-2018	0.25-in. Drill bit	211	NeverFail, Inc.	8	\$2.73
211347	RU-995748G	15-Jan-2018	Band saw	309	BeGood, Inc.	1	\$31.59
211348	AA-E3522QW	15-Jan-2018	Rotary sander	211	NeverFail, Inc.	2	\$34.46
211349	GH-778345P	16-Jan-2018	Power drill	157	ToughGo, Inc.	1	\$69.32

6. Using the answer to the above question, remove all partial dependencies and draw the new dependency diagrams.

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